Week 12: NoSQL and MongoDB CRUD

Learning outcomes: CLO1, CLO5

Lesson

- NoSQL types:
- **document** (MongoDB): records with varying fields, e.g. product catalogues;
- **key-value** (Redis): fast lookups, e.g. caches and sessions;
- **wide-column** (Cassandra): huge write-heavy data, e.g. sensor logs;
- **graph** (Neo4j): connections, e.g. friend networks.
- Choose by how the data is used.
- **CAP:** during a network partition, a distributed database must choose consistency or availability.
- MongoDB stores **documents** (BSON) in **collections**. Fields can be objects or arrays; documents can differ.
- `find(filter, projection)`. Operators: `$gt`, `$in`, … Dot notation reaches nested fields: `"districts.district_pcode"`.
- Read `cambodia`; write `student_work`. `_id` is added by MongoDB.

Worked example

Run `example.js` (how (`setup/README.md`)), then answer in `work.js`.

Collections for this week

Same rows and field names as MySQL, one document per row, plus `_id`. Read `cambodia`, write `student_work`.

Collection	Field	Type	Meaning
`province`	`province_pcode`	string	Province code
`province`	`province_name`	string	Province name (English)
`province`	`area_km2`	number	Area in km², one decimal place
`district`	`district_pcode`	string	District code
`district`	`district_name`	string	District name (English)
`district`	`area_km2`	number	Area in km², one decimal place
`district`	`province_pcode`	string	Province code
`commune`	`commune_pcode`	string	Commune code
`commune`	`commune_name`	string	Commune name (English)
`commune`	`area_km2`	number	Area in km², one decimal place
`commune`	`district_pcode`	string	District code

Links: `district.province_pcode` → `province`; `commune.district_pcode` → `district`. MongoDB does not enforce them.

Your collection: `example.js` copies provinces to `student_work.w12_province` with a unique index on `province_pcode`.

More: data guide (`data/README.md`).

Lab tasks

1. Run `example.js`; look at one document per collection.
2. Find provinces above an area you choose; show code, name and area only.
3. Copy provinces to `student_work.w12_province` with a unique index on `province_pcode`.
4. Delete one document, insert it again, check its fields against `cambodia.province`.
5. Update one name to its current value. `matchedCount` vs `modifiedCount`?

Homework

1. Write four CRUD commands with output; compare each with SQL.
2. When would embedding districts in provinces help? When is SQL better?
3. In `student_work`, build one province document with an array of its districts. Query `districts.district_pcode` and use `$size` or `$elemMatch`.

Capstone sprint

C12: Add a document part to your project and map your tables to documents.

Submit

Answers in `work.js`, reflection in `reflection.md`, output in `results/`. Commit and push. See SUBMISSIONS.md.